

Initialize Boundaries: $LB = -\infty$, $UB = +\infty$
Set Iteration Counter: $k = 1$
Empty Set of Worst-case Scenarios: $\mathcal{U}^* = \emptyset$

Master Problem (Minimization)

1. Optimize Day-Ahead strategy (P^{DA}) given known $\mathcal{U}^*$.
2. Objective gives the overall best-case cost.
3. Update Lower Bound: $LB = Obj(MP)$

Pass fixed Day-Ahead purchases (P^{DA})

Subproblem (Maximization)

1. Given fixed P^{DA} , find the worst-case weather (ξ^*).
2. $Obj(SP) = \text{Maximum Real-time balancing penalty}$.
3. Update Upper Bound: $UB = \min(UB, Cost^{DA} + Obj(SP))$

Pass updated UB

Convergence?

$$\frac{UB-LB}{LB} \leq \epsilon$$

YES

STOP

Optimal Robust
Strategy Found!

C&CG Cut Generation

1. Save worst-case ξ^* to $\mathcal{U}^*$.
2. Create new real-time variables $\mathcal{Y}^{(k+1)}$.
3. Add operational constraints to MP.
4. $k = k + 1$

*Feedback generated
C&CG constraints to MP*

NO